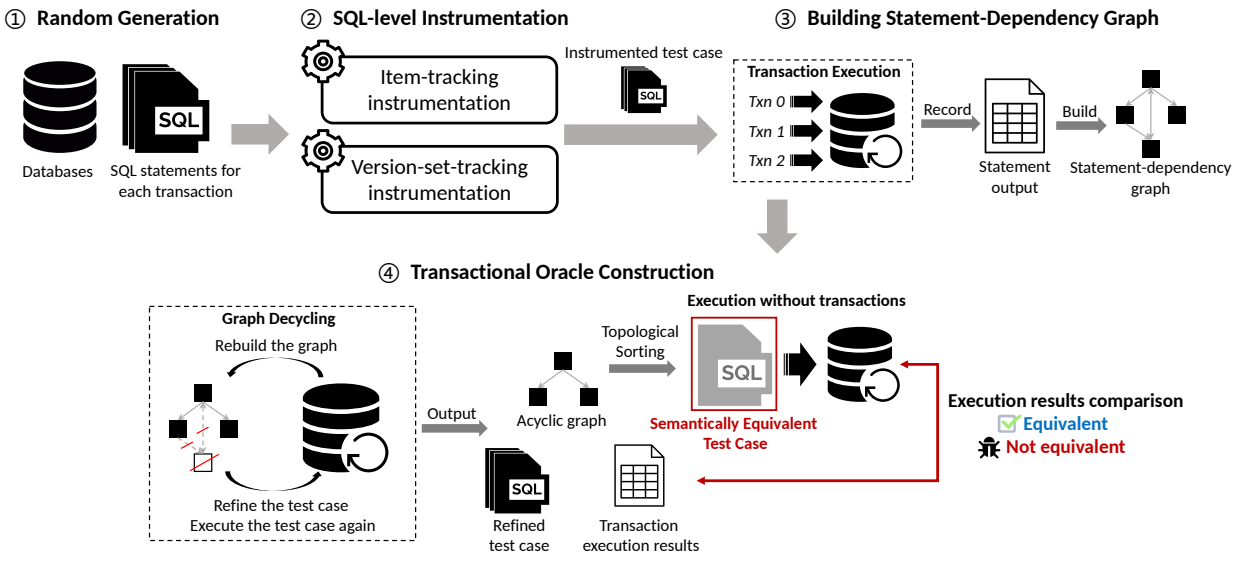
① Random Generation
② SQL-level Instrumentation
③ Building Statement-Dependency Graph
Databases
SQL
SQL statements for each transaction
Item-tracking instrumentation
Version-set-tracking instrumentation
Instrumented test case
SQL
Transaction Execution
Txn 0
Txn 1
Txn 2
Record
Statement output
Build
Statement-dependency graph
④ Transactional Oracle Construction
Graph Decycling
Rebuild the graph
Refine the test case
Execute the test case again
Output
Acyclic graph
Topological Sorting
Execution without transactions
SQL
Semantically Equivalent Test Case
Execution results comparison
Equivalent
Not equivalent
SQL
Refined test case
Transaction execution results